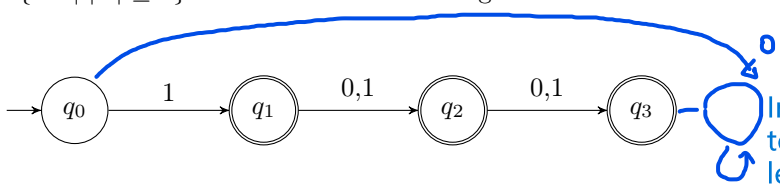
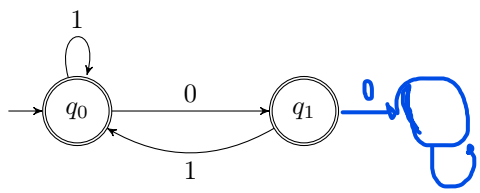
Exercise 3.1

(a) The language $L_1 = \{1w \mid |w| \leq 2\}$ consists of words starting with '1' with a maximum length of 3.



In a DFA every state has to take input from every letter from the alphabet (0,1 in our case

(b) The language $L_2 = \{w \in \{0,1\}^* \mid w \text{ does not contain two consecutive 0s}\}$.



1.5/2

Exercise 3.2

(a) The language $L(M)$ consists of words over the alphabet $\Sigma = \{a, b, c\}$ that must contain at least one symbol a. Each sequence of a's can be followed by an optional b and any number of c's including 0. Due to ϵ-transitions, the b can be omitted and the process can be repeated, but the word must always begin with at least one a to reach an accepting configuration. ✓

(b) The ϵ-closures $E(q)$ for each state q of the NFA M are as follows:

- $E(q_0) = \{q_0\}$
- $E(q_1) = \{q_0, q_1\}$
- $E(q_2) = \{q_0, q_1, q_2\}$ ✓

(c) To show that the NFA M accepts the word $w = abac$, we specify the sequence of states $(q'_0, q'_1, q'_2, q'_3, q'_4)$ as follows:

$$q'_0 = q_0, \quad q'_1 = q_2, \quad q'_2 = q_1, \quad q'_3 = q_2, \quad q'_4 = q_1$$ ✓

This sequence satisfies the properties required by the definition on slide 27:

1. $q'_0 = q_0 \in E(q_0) = \{q_0\}$.
2. $q'_1 = q_2 \in \bigcup_{q \in \delta(q'_0, a)} E(q)$, because $\delta(q_0, a) = \{q_2\}$ and $q_2 \in E(q_2)$.
3. $q'_2 = q_1 \in \bigcup_{q \in \delta(q'_1, b)} E(q)$, because $\delta(q_2, b) = \{q_1\}$ and $q_1 \in E(q_1)$.
4. $q'_3 = q_2 \in \bigcup_{q \in \delta(p,a), p \in E(q'_2)} E(q)$, because $q_0 \in E(q_1)$, $\delta(q_0, a) = \{q_2\}$ and $q_2 \in E(q_2)$.
5. $q'_4 = q_1 \in \bigcup_{q \in \delta(p,c), p \in E(q'_3)} E(q)$, because $q_1 \in E(q_2)$, $\delta(q_1, c) = \{q_1\}$ and $q_1 \in E(q_1)$.
6. $q'_4 = q_1 \in F$, where $F = \{q_1\}$.

The sequence $(q_0, q_2, q_1, q_2, q_1)$ proves that $w = abac$ is accepted by M.

(d) Given NFA $M = \langle Q, \Sigma, \delta, q_0, F \rangle$

where, Q $= \{q_0, q_1, q_2\}$

$\Sigma = \{a, b, c\}$

$F = \{q_1\}$

We can construct a DFA $N = \langle Q', \Sigma, \delta', q_0', F' \rangle$ with $L(M) = L(N)$. Here N is defined as follows:

$Q' = \mathcal{P}(Q) = \{\emptyset, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$

$q_0' := E(q_0)$ where ε closure of q_0 is $\{q_0\}$

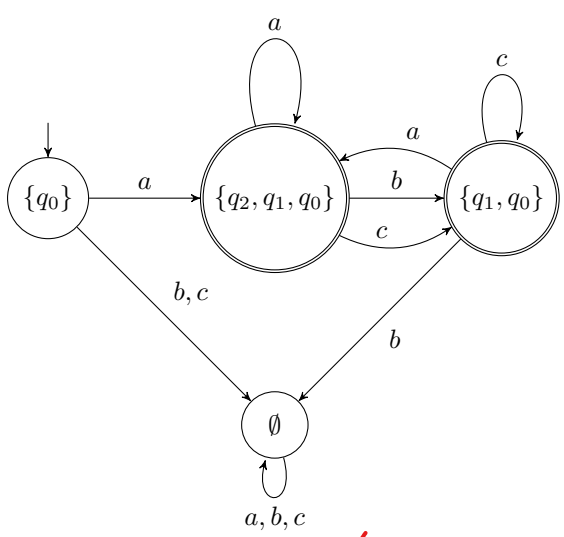
$F' := \{\mathcal{Q} \subseteq Q | \mathcal{Q} \cap F \neq \emptyset\}$

Hence, Accepting states F' are those states where the intersection with the Accepting state F of the NFA is not zero. In other words, any state which has q_1 is an accepting state; thus $\{q_1\}, \{q_0, q_1\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$

For all $\mathcal{Q} \in Q' : \delta'(\mathcal{Q}, a) := \cup_{q \in \mathcal{Q}} \cup_{q' \in \delta(q,a)} E(q')$

- We start from $\{q_0\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. State q_0 on b, c goes to $\emptyset$.
- State $\{q_1\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. Taking c goes to q_1 again and b goes to $\emptyset$.
- State $\{q_2\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. Taking b goes to $\{q_1, q_0\}$ and c goes to $\emptyset$.
- State $\{q_0, q_1\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. Taking c gives $\{q_1, q_0\}$ and b goes to $\emptyset$.
- State $\{q_0, q_2\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. Taking c gives $\{q_1, q_0\}$ and b goes to $\emptyset$.
- State $\{q_1, q_2\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. Taking b gives $\{q_1, q_0\}$ and c goes to $\emptyset$.
- State $\{q_0, q_1, q_2\}$ - Taking a gives state $\{q_2, q_1, q_0\}$. Taking b, c goes to $\{q_1, q_0\}$.

We may simplify this by observing that no arrows point at states $\{q_1\}, \{q_2\}, \{q_0, q_2\}$ and $\{q_1, q_2\}$, so they may be removed without affecting the performance of the machine. Doing so yields the following DFA N:



3.5/4

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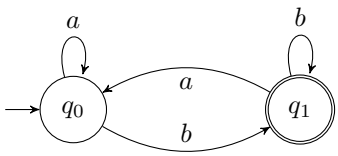
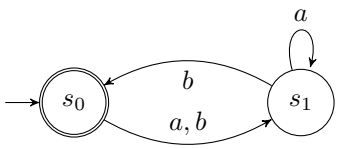
Exercise 3.3

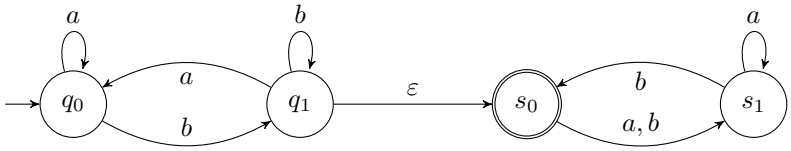
(a) M1:



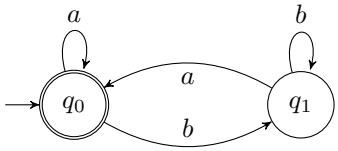
M2:



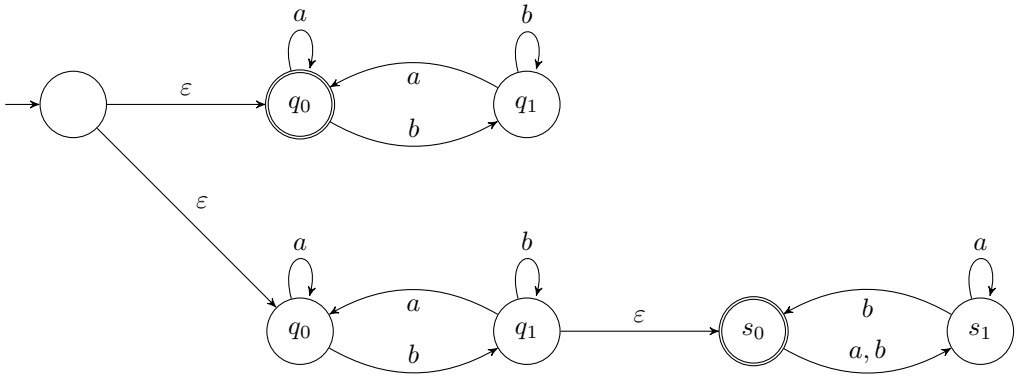
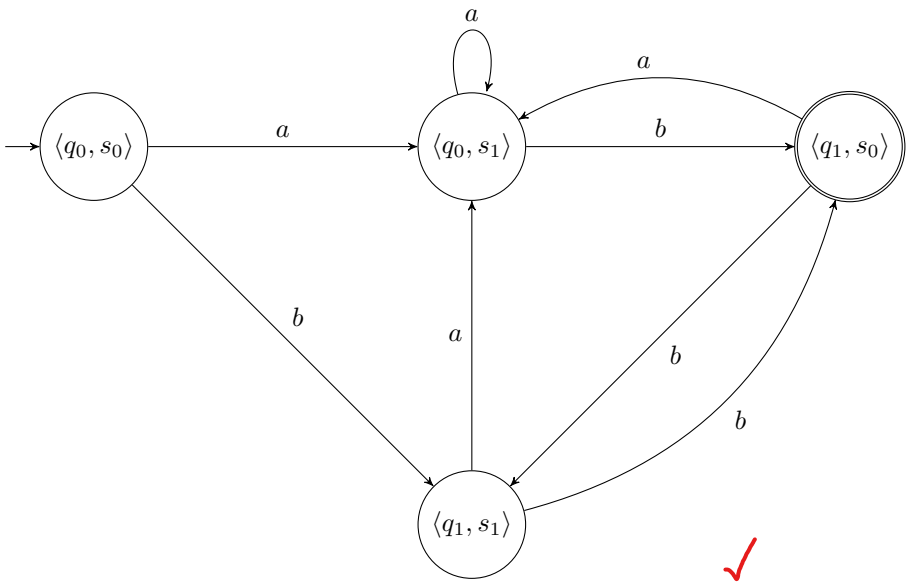
$\mathcal{L}(M_1)\mathcal{L}(M_2)$:



$\overline{\mathcal{L}(\mathrm{M}_1)}$:



$\overline{\mathcal{L}(\mathrm{M}_1)} \cup \mathcal{L}(M_1)\mathcal{L}(M_2)$:



(b) $\mathcal{L}(M_1) \cap \mathcal{L}(M_2)$: